Sagar Kriplani

Mechatronic & Automation Engineer | Semiconductor Precision Mechanics & DFM/DFA R&D Engineer

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EU Blue Card (Unrestricted)

PROFESSIONAL SUMMARY & CORE STRENGTHS

Mechatronic and Development Engineer holding an M.Sc. in Scientific Instrumentation (GPA 1.8, Ernst-Abbe-Hochschule Jena) and a Bachelor in Mechanical Engineering. Multi-year industry background in custom automated machinery (8+ platforms), sensor-actuator integration, and cross-functional coordination between mechanical, electrical, and control disciplines. Experienced in DIN/ISO, VDE, and CE machinery compliance, structured troubleshooting, and commissioning.

PROFESSIONAL EXPERIENCE

Mechanical & Mechatronic Design Engineer — NeSt Maschinen- und Anlagentechnik GmbH 09/2021 – 12/2022 | Roding, Germany

- **Special Machinery Engineering:** Directed mechanical development and series release of **8+ custom machinery platforms**; closely collaborated with electrical engineers on control cabinet integration and cable routing.
- **Mechatronic Interface Design:** Sized and integrated sensors (optical, proximity, pressure), pneumatic actuators, servo motors, and safety interlocks complying with CE and VDE machinery standards.
- **Standards Compliance:** Issued ~200 manufacturing and assembly drawings per month strictly adhering to **GD&T (DIN EN ISO 1101)**.
- **Assembly & Commissioning Support:** Supported shop-floor assembly, resolved electro-mechanical interfacing discrepancies, and maintained engineering change orders (ECO).

Master Thesis Candidate & R&D Intern — Brooks Automation (Germany) GmbH 09/2023 – 10/2024 | Jena, Germany

- **Sensors & Automated Testing:** Instrumented automated test rigs with thermal, pressure, and gas flow sensors; performed systematic failure mode investigations.
- **Thermo-Mechanical Simulation:** Performed coupled thermal-structural FEA in ANSYS Workbench to verify operational integrity.

ADDITIONAL ENGINEERING EXPERIENCE

Design Engineer (Electromechanical Products & Tooling) — Ravi Brass Industries 08/2019 – 08/2021 | Jamnagar, India

- **Electromechanical Components:** 3D CAD design of precision electrical terminals, connectors, and switchgear components.
- **Tooling Design:** Maintained 4 injection molding/stamping dies supporting >100,000 components/month.

Research Assistant / Mechatronic Testing — Fraunhofer IOF 03/2023 – 08/2023 | Jena, Germany

- **Sensor Test Fixtures:** Assembled and calibrated mechatronic test fixtures for optical sensors.

EDUCATION & ACADEMIC CREDENTIALS

Master of Science (M.Sc.) — Scientific Instrumentation 10/2021 – 10/2024

Ernst-Abbe-Hochschule Jena, Germany | Final Grade: 1.8 (ECTS A)

Focus: Sensor Systems, Actuators, Metrology, Materials Science, and Structural FEA. Master thesis completed at Brooks Automation (Grade: 1.3).

Bachelor of Engineering (B.Eng.) — Mechanical Engineering 08/2015 – 06/2019

Gujarat Technological University, India | Final Grade: 1.9 (8.12/10 CGPA)

CORE SKILLS & TECHNICAL COMPETENCIES

Custom Machinery:	Mechatronic assembly, sensor-actuator integration, pneumatics, cabling paths, DFM/DFA
Codes & Standards:	CE Machinery Directives, VDE/EN safety requirements, **GD&T (DIN EN ISO 1101)**, FMEA
CAD & Simulation:	SolidWorks, Autodesk Inventor, PTC Creo, CAE basics, ANSYS Workbench (Certified 325/325), MATLAB, Python
Testing & Diagnostics:	Commissioning, test bench instrumentation, fault isolation, calibration, FLIR thermography
Languages:	**German** (B2/C1 Professional), **English** (C1 Fluent)
Location & Mobility:	Resident in 89518 Heidenheim (18 km / ~18 min to Niederstotzingen), **EU Blue Card**, immediately available